Yishen Song 宋义深

Ph.D. Student, Beijing Normal University, Beijing, China

songyishen@outlook.com | ORCID: 0009-0004-4846-7635 | http://eason0212.com/

RESEARCH INTERESTS

Artificial Intelligence and Educational Measurement, Computerized Adaptive Testing, STEM Education

EDUCATION

Beijing Normal University, Beijing, China — Sept. 2024 –Present

Ph.D. Student in Internet Education

Beijing Normal University, Beijing, China — Sept. 2022 –Aug. 2024

Master Student in Distance Education (transferred to Ph.D. program in Aug. 2024)

Beijing Normal University, Beijing, China — Sept. 2018 –Jul. 2022

Bachelor of Science in Educational Technology

PUBLICATIONS

Journal papers (in English)

- **Song, Y.**, Li, H., & Zheng, Q. (2026). Multi-agent automated item generation based on large language models. *Educational Technology & Society*, 29(2), 233–248, `https://doi.org/10.30191/ETS.202604_29(2).SP02`
- Du, J., **Song, Y.**, & Zheng, Q. (2026). LLM-simulated nonequivalent groups with anchor test: a novel approach for test equating in the absence of traditional anchor items. *IEEE Transactions on Learning Technologies*, 19, 75–86, `https://doi.org/10.1109/TLT.2026.3661154`
- **Song, Y.**, Du, J., & Zheng, Q. (2025). Automatic item generation for educational assessments: a systematic literature review. *Interactive Learning Environments*, 33(9), 5386-5405, `https://doi.org/10.1080/10494820.2025.2482588`
- **Song, Y.**, Guo, L. & Zheng, Q. (2025). Measuring scientific inquiry ability related to hands-on practice: an automated approach based on multimodal data analysis. *Education and Information Technologies*, 30, 4381–4411, `https://doi.org/10.1007/s10639-024-12991-7`
- **Song, Y.**, Zhu, Q., Wang, H., & Zheng, Q. (2024). Automated essay scoring and revising based on open-source large language models. *IEEE Transactions on Learning Technologies*, 17, 1880-1890, `https://doi.org/10.1109/TLT.2024.3396873`

Journal papers (in Chinese)

- 郑勤华, **宋义深**, 李爽. (2026). 中小学生智能素养的表现性评价：工具设计、数据表征与智能分析. *中国电化教育*, (01), 92–98.
- 杜君磊, 郑勤华, **宋义深**. (2025). 大语言模型在无锚题等值中的应用：以阅读素养测评为例. *武汉大学学报 (理学版)*, 71(05), 591–598. `https://doi.org/10.14188/j.1671-8836.2024.0088`
- 郑勤华, 刘司卓, **宋义深**. (2024). 政策视角下综合素质评价的定位与实施路径研究. *中国电化教育*, (12), 81–89.
- **宋义深**. (2024). 多模态教育数据在教育实证研究中的应用——基于国际实证研究的系统性文献综述. *开放学习研究*, 29(03), 45–52. `https://doi.org/10.19605/j.cnki.kfxxyj.2024.03.005`
- 郑勤华, **宋义深**. (2024). 互联网教育的发展模式研究——基于利益演化博弈的仿真视角. *中国远程教育*, 44(05), 35–50. `https://doi.org/10.13541/j.cnki.chinade.2024.05.007`
- **宋义深**, 冯晓英. (2023). 在线学习失信行为的驱动因素与形成机制研究——基于扎根理论. *电化教育研究*, 44(09), 64–70. `https://doi.org/10.13811/j.cnki.eer.2023.09.009`
- 傅骞, **宋义深**, 郑娅峰. (2023). 面向信息科技教育的物联网平台研究——MixIO 物联网平台的设计与实现. *电化教育研究*, 44(05), 99–103+128. `https://doi.org/10.13811/j.cnki.eer.2023.05.013`

Conference papers

- **Song, Y.** (2025). CoScore: an LLM-empowered human-AI collaborative text scoring rule generation and calculation engine. *IEEE International Conference on Teaching, Assessment, and Learning for Engineering (IEEE TALE)*, `https://doi.org/10.1109/TALE66047.2025.11346772`

PROJECTS

MixIO: An education-first open-source IoT platform Beijing, China
Main Contributor Aug. 2020 –Present

- Developed the front-end and back-end of an IoT Server for K-12 IT education.
- Maintained an open source repository: `https://gitee.com/mixly2/mixio`
- The official server has around 1,000 active users per month, supporting students in building their own IoT Apps in a low-code manner: `https://mixio.mixly.cn/`

EXPERIENCE

Intern Information Technology Teacher Beijing, China
Beijing Xicheng Foreign Languages School (BFLS) Oct. 2021 –Dec. 2021

- Taught Python programming to first-year high school students.
- Designed teaching activities and produced multimedia teaching materials.

AWARDS

- **National Scholarship for Doctoral Students of China** Dec. 2024
  Ministry of Education, China
- **National Scholarship for Undergraduate Students of China** Dec. 2021
  Ministry of Education, China

SKILLS

- **IELTS (Academic): 7.5** (Listening: 8.0, Reading: 8.5, Writing: 6.5, Speaking: 6.0)
- Python (proficient), JavaScript (proficient), R (familiar), Stata (familiar)